

ASSIGNMENT #4

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
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Your name: _____

Due Date: Monday March 09 by 6:00pm in my office

Number of points.....100

Your score.....

Note: All problems are worth 100 points. Your final score will be the average value

Problem 4A: Acentric Factor for Redlich-Kwong Equation of State

Calculate the numerical value of the acentric factor ω for all fluids that follow the Redlich-Kwong Equation of State given below:

$$P + \frac{a}{V(V+b)\sqrt{T_r}} = \frac{RT}{V-b} \quad \text{where:} \quad a = 0.42748 \frac{(RT_c)^2}{P_c} \quad b = 0.08664 \frac{RT_c}{P_c}$$

Problem 4B: Applications of the Peng-Robinson Equation of State

The following table show the isobaric properties of benzene at pressure of 5 bar. Using the Peng-Robinson equation of state estimate these properties and calculate the errors of your predictions and their average value for each property.

Isobaric Properties of Benzene at P=5 bar							
Phase	Temperature (deg C)	Pressure (bar)	Enthalpy (kJ/mol)	Cv kJ/(kmol K)	Cp kJ/(kmol K)	Sound Speed (m/s)	Joule-Thomson (K/bar)
Liquid	50	5	-4.2864	97.389	138.74	1192.0	-0.038923
Liquid	60	5	-2.8846	100.06	141.64	1144.0	-0.037061
Liquid	70	5	-1.4535	102.86	144.58	1096.4	-0.035153
Liquid	80	5	0.007071	105.68	147.53	1049.4	-0.033185
Liquid	90	5	1.4969	108.44	150.44	1003.4	-0.031128
Liquid	100	5	3.0158	111.12	153.33	958.31	-0.028942
Liquid	110	5	4.5636	113.71	156.22	914.07	-0.026582
Liquid	120	5	6.1404	116.21	159.14	870.53	-0.023992
Liquid	130	5	7.7466	118.63	162.12	827.54	-0.021102
Liquid	140	5	9.3832	120.99	165.21	784.86	-0.017830
Liquid	142.61	5	9.8148	121.60	166.04	773.77	-0.016902
Vapor	142.61	5	36.703	111.53	125.35	198.87	2.2072
Vapor	150	5	37.634	113.19	126.46	201.79	2.0381
Vapor	160	5	38.906	115.43	128.10	205.56	1.8393
Vapor	170	5	40.196	117.68	129.84	209.13	1.6687
Vapor	180	5	41.504	119.92	131.67	212.55	1.5211
Vapor	190	5	42.830	122.16	133.55	215.84	1.3926
Vapor	200	5	44.175	124.38	135.47	219.00	1.2800

Note: Enthalpy = 0.0 kJ/mol at the normal boiling point for saturated liquid Benzene.

List of useful formulas and hints to help you complete this assignment:

- The Peng-Robinson equation of state and its various standard forms:

$$P + \frac{a\sigma}{V(V+b)+b(V-b)} = \frac{RT}{V-b} \quad \left| \quad \frac{d\sigma}{dT} = -\frac{1}{T_c} \frac{\kappa\sqrt{\sigma}}{\sqrt{T_r}} \right.$$

$$Z = \frac{1}{1-b\rho} - \frac{a\sigma}{RT} \frac{\rho}{(1+b\rho)+b\rho(1-b\rho)} \quad \left| \quad \frac{d^2\sigma}{dT^2} = \frac{\kappa+1}{2T_c^2} \frac{\kappa}{T_r\sqrt{T_r}} \right.$$

- The residual heat capacity function:

$$C_p^R = -R - T \int_V^\infty \frac{\partial^2 P}{\partial T^2} dV + (C_p - C_v)$$

- The difference between the two heat capacities:

$$C_p - C_v = T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P = -T \frac{\left[\left(\frac{\partial P}{\partial T} \right)_V \right]^2}{\left(\frac{\partial P}{\partial V} \right)_T}$$

- The speed of sound ν and Joule-Thomson coefficient μ

$$\nu^2 = \gamma \left(\frac{\partial P}{\partial \rho} \right)_T = -\gamma V^2 \left(\frac{\partial P}{\partial V} \right)_T \quad \text{where: } \gamma = \frac{C_p}{C_v} \quad \left| \quad \mu = \left(\frac{\partial T}{\partial P} \right)_H = \frac{\beta T - 1}{\rho C_p} \quad \text{where: } \beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

- **Hint #1:** Derive an expression for:

$$\rho \left(\frac{\partial Z}{\partial \rho} \right)_T \quad \text{or} \quad \frac{V}{P} \left(\frac{\partial P}{\partial V} \right)_T = \text{function}(Z, A, B)$$

- **Hint #2:** Derive an expression for:

$$T \left(\frac{\partial Z}{\partial T} \right)_\rho \quad \text{or} \quad \frac{T}{P} \left(\frac{\partial P}{\partial T} \right)_V = \text{function}(Z, A, B, T_r)$$

- **Hint #3:** Develop formulas that only depend on $\{Z, A, B, T_r\}$ for all quantities of interest of your assignment. That way you can implement them in Excel and perform your calculations.

Place all your answers in boxes for easy readability. Clearly state your assumptions and references for additional data you may need to answer the questions.